

Total No. of Questions : 8]

SEAT No. :

PC-4995

[Total No. of Pages : 2

[6354]-853

B.E. (Artificial Intelligence and Machine Learning)
HONOURS IN SOFT COMPUTING AND DEEP LEARNING
(2019 Pattern) (Semester - VIII) (410303)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Explain how genetic algorithm are different from evolutionary programming? [6]
b) What are operators in genetic algorithm? List and explain GA operators in brief? [6]
c) Explain Vncoding methods? [6]

OR

- Q2)** a) What are types of crossover and mutation techniques? [6]
b) Mention application area of genetic algorithm? [6]
c) What is difference between genetic algorithm and genetic programming? [6]

- Q3)** a) List out the strength and weakness of artificial neural network? [6]
b) Explain in brief architecture of multilayer feed-forward neural network? [6]
c) Explain how weights are updated in perceptron network? [5]

OR

- Q4)** a) Explain Types of Artificial Neural Network (ANN). [6]
b) Explain Single layer and multilayer Perceptrons. [6]
c) Explain Self-Organizing Map in neural network? [5]

P.T.O.

- Q5)** a) Explain why to use deep learning? [6]
b) Explain in brief architecture of deep network? [6]
c) Explain different Deep Learning libraries? [6]

OR

- Q6)** a) Explain Restricted Boltzman Machines in details? [6]
b) Explain different Deep Learning platform. [6]
c) Explain any one Use Cases of Deep Learning? [6]

- Q7)** a) Explain in detail Convolution Neural Network(CNN)? [6]
b) Explain different Properties of CNN representations? [6]
c) List and explain different Applications of CNN? [5]

OR

- Q8)** a) Explain in details Recurrent Neural Network (RNN)? [6]
b) Explain Sequential processing LSTM model in details? [6]
c) Explain difference between CNN and RNN? [5]

